

AIM:

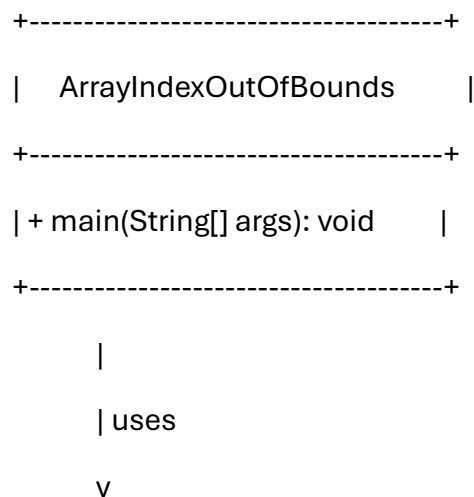
To write, compile, and execute a Java program to demonstrate the concept of exception handling using the try-catch mechanism, specifically for handling an `ArrayIndexOutOfBoundsException`. The program accesses an invalid array index intentionally, catches the exception using the appropriate catch block, displays a meaningful error message, and allows the remaining statements of the program to execute normally without abnormal termination.

Algorithm:

ALGORITHM

1. **Start** the program.
2. Define a class named `ArrayIndexOutOfBoundsException`.
3. Declare and initialize an integer array with three elements: 10, 20, 30.
4. Use a **try block** to access an element of the array.
5. Attempt to access `numbers[5]`.
6. Since the array contains elements only at indexes 0, 1, and 2, accessing index 5 generates an `ArrayIndexOutOfBoundsException`.
7. Transfer the control to the **catch block**.
8. Catch the exception using `ArrayIndexOutOfBoundsException`.
9. Display the exception message using the exception object.
10. Execute the statement "**Program continues....**" after the catch block.
11. **Stop** the program.

UML Diagram:



```

+-----+
| ArrayIndexOutOfBoundsException |
+-----+
| + getMessage(): String      |
| + toString(): String        |
+-----+

```

Code:

```

package Observations;

public class ArrayIndexOutOfBoundsException {
    public static void main(String[] args) {
        int[] numbers = {10,20,30};
        try{
            System.out.println("element:" + numbers[5]);
        }catch(ArrayIndexOutOfBoundsException e){
            System.out.println("expection caught: " + e);

        }
        System.out.println("program continues....");
    }
}

```

Output:

```

expection caught: java.lang.ArrayIndexOutOfBoundsException: Index 5 out of bounds for length 3
program continues....
PS C:\Users\Sahithi pyla\Desktop\Sahithi java>

```